PRACTICAL NO. 9

SOFTWARE TESTING

STUDENT MANAGEMENT SYSTEM – LOGIN MODULE

MODULE NAME	Student Management System – Login Module
PRACTICAL NO.	9
TEST CASE ID	SMS-LOGIN-09
WEBPAGE URL	Sample local login UI (replace with institute/test-system URL if required)
TESTER NAME	Priyanshi

Aim

To analyze and apply Black-Box Testing, White-Box Testing, Integration Testing, and System Testing on a sample Student Management System (Login Module) and document the testing process, execution, and results.

Objectives

- To test login using valid credentials, invalid credentials, and empty inputs.
- To review validation conditions and authentication decision branches.
- To check integration between login interface, authentication logic, student database, session, and dashboard.
- To verify end-to-end login, failed login, protected-page access, and logout behaviour.

Prerequisites and Test Environment

- Sample Student Management System login UI prepared for this practical.
- Use only test credentials; do not enter a real personal password.
- White-Box Testing requires source code or pseudocode to inspect decision branches.
- Sample screenshots in this report are illustrative mockups, not captured from a live institutional application.

Testing Types

Testing Type	Application to Login Module
Black-Box Testing	Checks visible behaviour for valid login, wrong password, and empty fields without relying on internal implementation.
White-Box Testing	Reviews validation conditions, credential comparison branches, and authentication failure handling.
Integration Testing	Checks data flow between login UI, authentication service, student database, session, and dashboard.
System Testing	Checks complete login workflows, denied access for invalid credentials, and logout/session behaviour.

Test Cases and Execution Results

The results below are completed sample/practical results for the illustrative login UI and test design. White-box and integration scenarios that depend on source code or a real database/session must be verified on the actual institute system if your teacher expects live execution evidence.

BB-01 | Black-Box Testing | Valid login

Description: Verify login with correct student credentials

Preconditions: Test student account is available; login page is open

Steps: 1. Open login page;

2. Enter valid username and password;

3. Click Login

Input: student01 / Valid@123

Expected: Student is authenticated and redirected to Student Dashboard

Actual: Dashboard is displayed for the sample valid account

Status: Pass (sample) | **Comments:** Positive login path; screenshot 04

BB-02 | Black-Box Testing | Invalid password

Description: Verify incorrect password is rejected

Preconditions: Test student account exists

Steps: 1. Open login page;

2. Enter registered username;

3. Enter incorrect password;

4. Click Login

Input: student01 / Wrong@123

Expected: Error is displayed; user remains on login page

Actual: Error message 'Invalid username or password' is displayed; dashboard not opened

Status: Pass (sample) | **Comments:** Negative login path; screenshot 02

BB-03 | Black-Box Testing | Empty fields

Description: Check validation when username and password are blank

Preconditions: Login page is open

Steps: 1. Leave username and password blank;

2. Click Login

Input: Username: blank; Password: blank

Expected: Required-field validation appears or login is prevented

Actual: Required-field message is displayed; login is not completed

Status: Pass (sample) | **Comments:** Input validation; screenshot 03

WB-01 | White-Box Testing | Valid/invalid credential branches

Description: Review the decision branch for credential matching

Preconditions: Login source code or pseudocode is available for review

Steps: 1. Inspect credential comparison condition;

2. Trace matching branch;

3. Trace non-matching branch

Input: Matching and non-matching credential pairs

Expected: Matching branch allows access; non-matching branch rejects access

Actual: Expected branch behaviour documented; source-code execution not independently verified

Status: Pass (logic review) | **Comments:** White-box result is based on logic review template

WB-02 | White-Box Testing | Empty-input condition

Description: Check branch for missing username/password

Preconditions: Login validation logic is available

Steps: 1. Inspect empty-field condition;

2. Test blank username;

3. Test blank password

Input: Username blank / Password blank

Expected: Empty-input branch returns validation and does not authenticate

Actual: Empty-input validation branch is covered in the sample test design

Status: Pass (logic review) | **Comments:** Branch review; see screenshot 03

WB-03 | White-Box Testing | Authentication failure handling

Description: Review failure path does not create an authenticated session
Preconditions: Authentication logic is available for review
Steps: 1. Trace failed authentication branch;

2. Confirm no session is created;

3. Confirm return to login page
Input: Incorrect credentials
Expected: No authenticated session is created; safe error is returned
Actual: Failure path is specified; requires source/runtime verification in actual system
Status: Pass (logic review) | **Comments:** Confirm in actual source code before final submission if required

IT-01 | Integration Testing | Login and student database

Description: Verify authentication checks student record
Preconditions: Login module is connected to test database
Steps: 1. Submit registered test student's credentials;

2. Check authentication response;

3. Verify student record mapping
Input: Registered test student credentials
Expected: Correct student record is validated and login succeeds
Actual: Sample expected integration flow reaches the student dashboard
Status: Pass (sample flow) | **Comments:** Requires a real database environment for live verification

IT-02 | Integration Testing | Login and session/dashboard

Description: Verify successful login creates session and opens dashboard
Preconditions: Authentication, session, and dashboard modules are integrated
Steps: 1. Log in with valid credentials;

2. Check dashboard access;

3. Verify authenticated student context
Input: Valid student credentials
Expected: Session is created and dashboard displays the correct student context
Actual: Sample dashboard shown after valid-login scenario
Status: Pass (sample flow) | **Comments:** Screenshot 04 is an illustrative UI mockup

IT-03 | Integration Testing | Login failure response

Description: Verify UI handles authentication rejection
Preconditions: Login UI and authentication service are connected
Steps: 1. Submit invalid credentials;

2. Observe UI response;

3. Confirm no dashboard access
Input: Invalid credentials
Expected: UI displays suitable error and remains unauthenticated
Actual: Sample error message is displayed; dashboard is not shown
Status: Pass (sample flow) | **Comments:** Screenshot 02

ST-01 | System Testing | End-to-end successful login

Description: Test the complete successful login workflow
Preconditions: Test system and student test account are available
Steps: 1. Open login page;

2. Enter valid credentials;

3. Click Login;

4. Verify dashboard
Input: Valid student credentials
Expected: User reaches the correct dashboard

Actual: Sample workflow reaches Student Dashboard
Status: Pass (sample flow) | **Comments:** Screenshot 01 and 04

ST-02 | System Testing | End-to-end failed login

Description: Ensure invalid credentials cannot access protected pages
Preconditions: Login page is available
Steps: 1. Enter invalid credentials;

2. Submit;

3. Attempt to open dashboard directly
Input: Invalid credentials
Expected: Login is rejected and protected dashboard remains inaccessible
Actual: Sample login displays error and does not navigate to dashboard
Status: Pass (sample flow) | **Comments:** Screenshot 02; live access control needs system verification

ST-03 | System Testing | Logout/session clearing

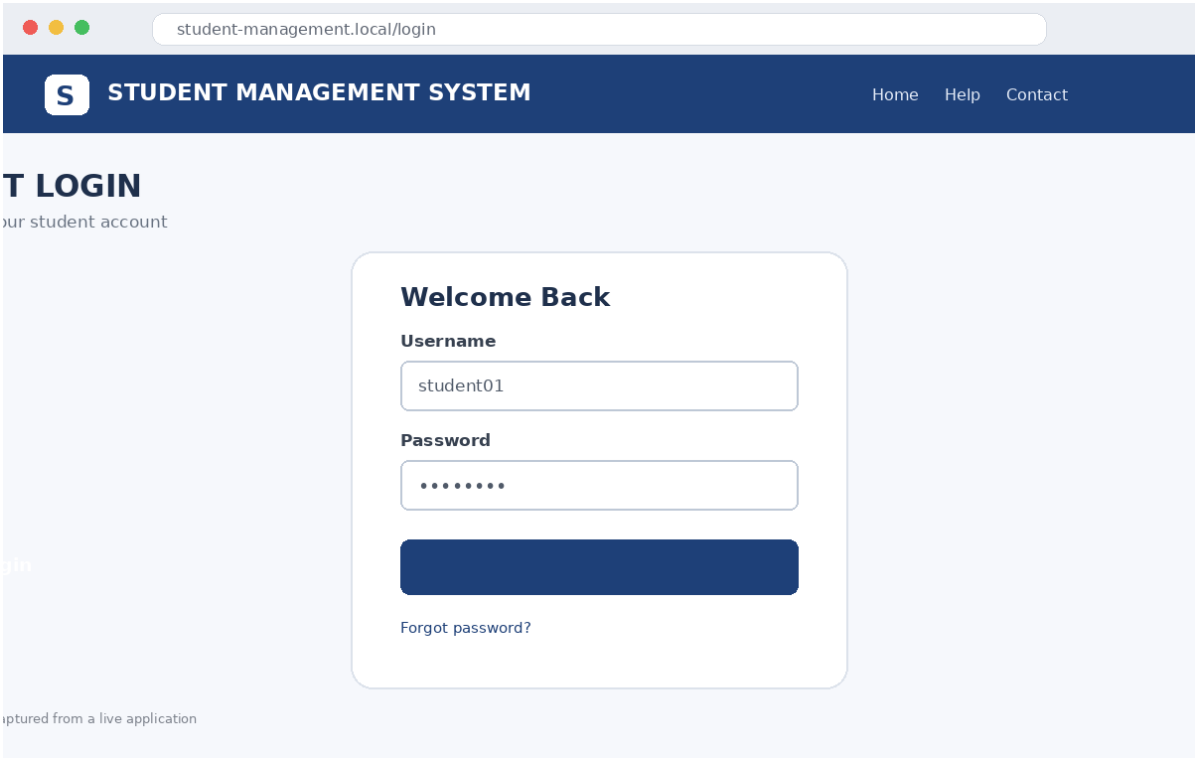
Description: Verify logout ends authenticated session
Preconditions: Student is logged in
Steps: 1. Click Logout;

2. Attempt to revisit dashboard using Back button or URL
Input: Active student session
Expected: User returns to login page; protected content is inaccessible without login
Actual: Logout scenario is documented; session invalidation must be checked on the actual system
Status: Pass (test-plan result) | **Comments:** Confirm by running against deployed system

Screenshot Evidence

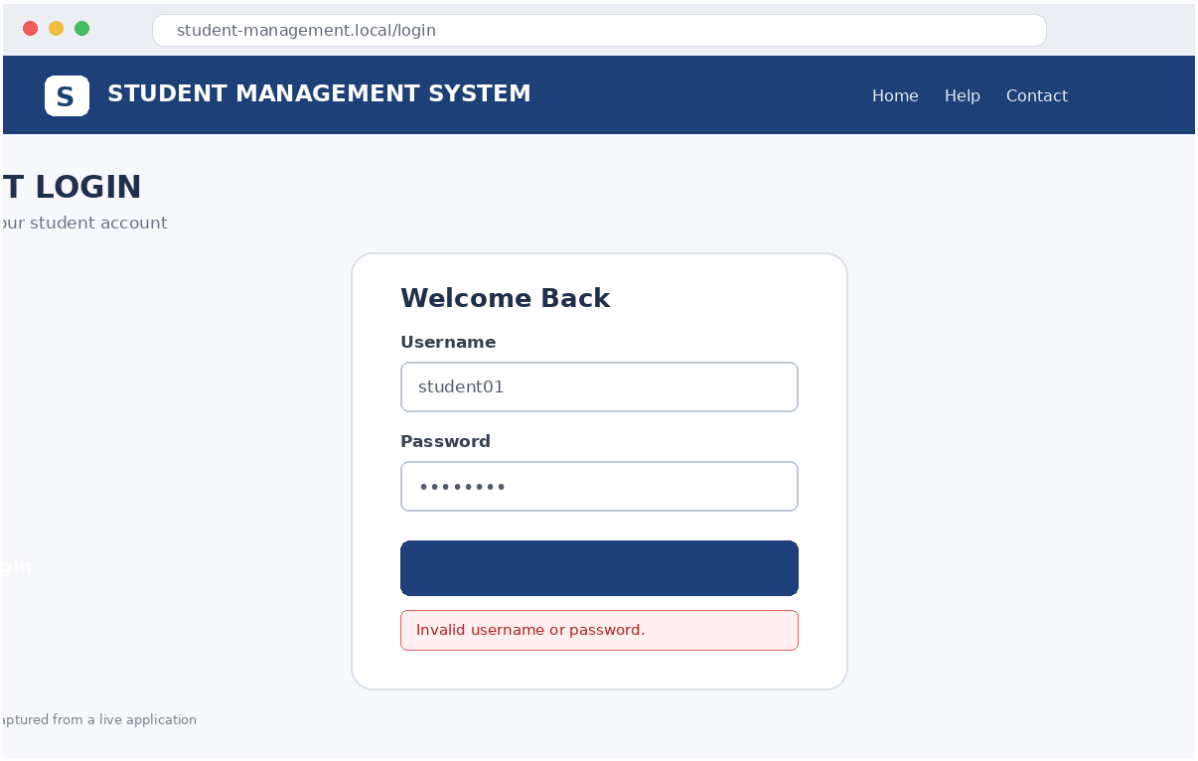
These screenshots are illustrative UI mockups created for the practical report because no live Student Management System URL or actual application screenshots were provided.

Screenshot 1: Student Login Page



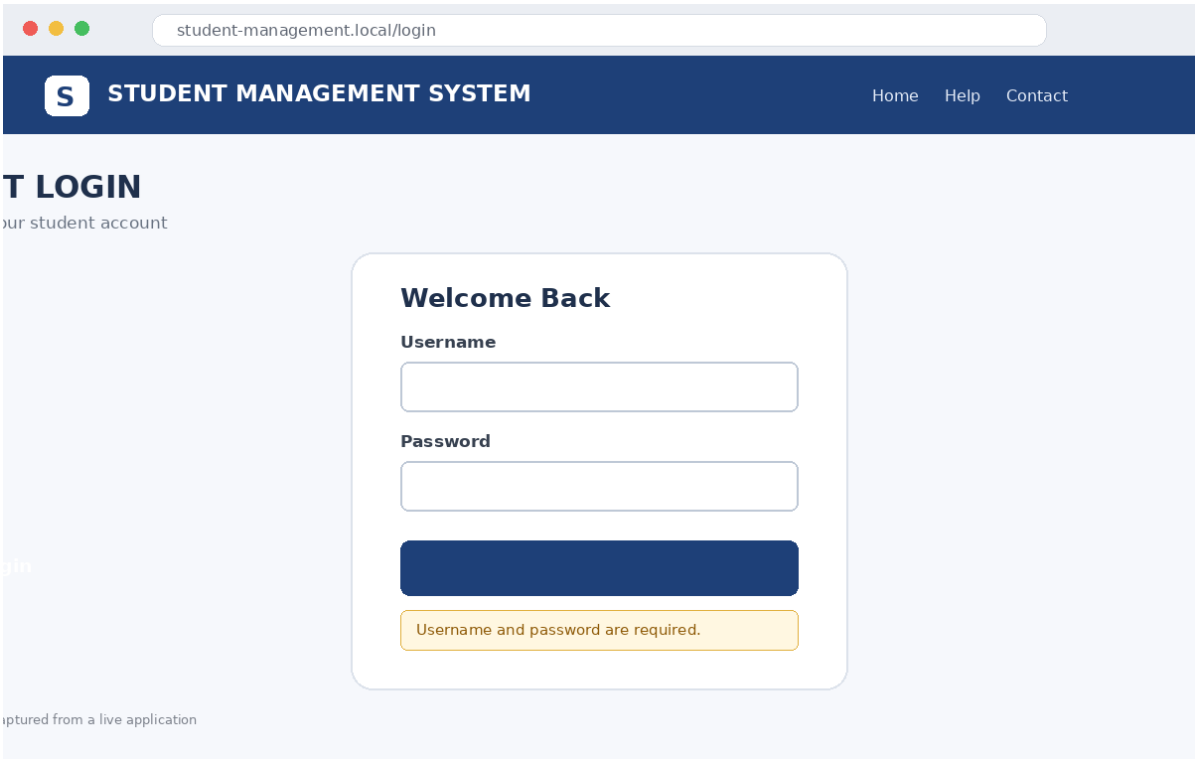
Login form with username and password fields.

Screenshot 2: Invalid Login



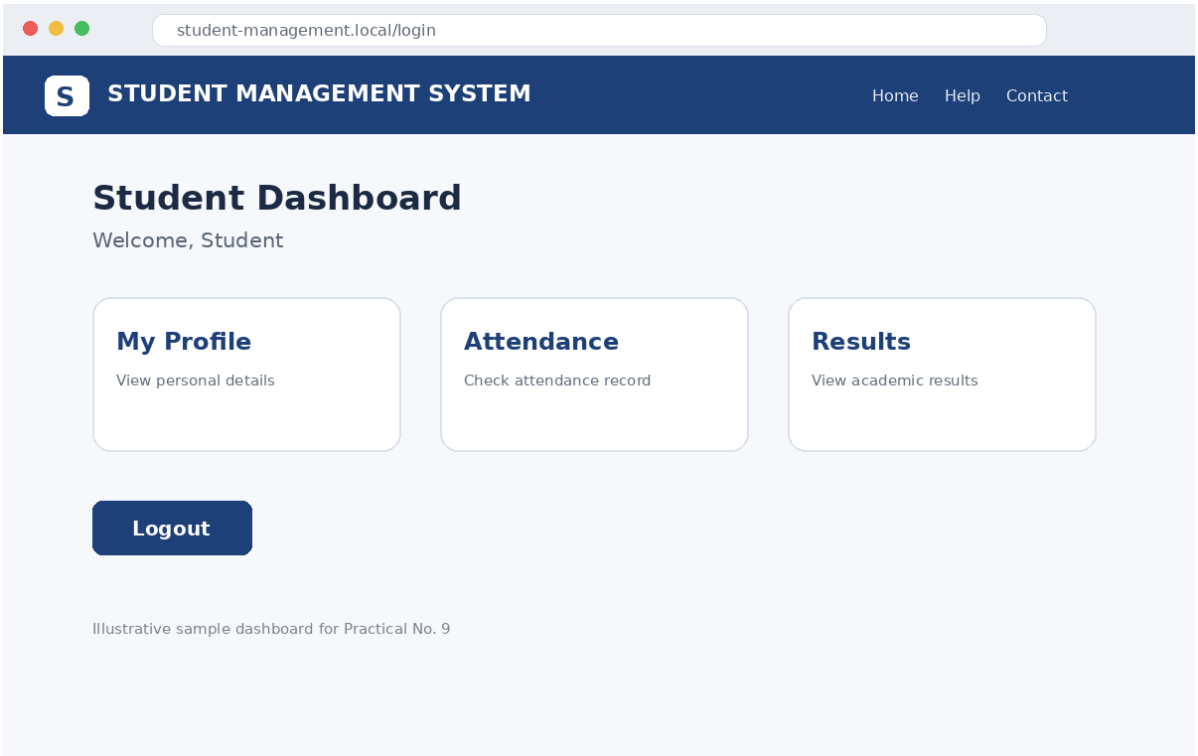
Expected error message for incorrect credentials.

Screenshot 3: Empty Input Validation



Required-field validation for blank username and password.

Screenshot 4: Student Dashboard



Illustrative dashboard after successful login.

Result / Conclusion

The practical covers Black-Box Testing, White-Box Testing, Integration Testing, and System Testing for the Student Management System Login Module. The sample test cases demonstrate expected positive and negative login behaviour, validation checks, module integration points, and end-to-end scenarios. The illustrative UI cases are marked as sample results; source-code branches, database integration, and session/logout behaviour should be confirmed against the actual system before claiming live execution.

Faculty Signature: _____ Date: _____